A/C LOW PRESSURE SWITCH

DESCRIPTION

The A/C low pressure switch is a single pole, single throw, pressure actuated switch that is used to control evaporator freeze up.

The A/C low pressure switch is mounted on a fitting located on the top of the A/C accumulator. The A/C low pressure switch fitting on the accumulator is equipped with an O-ring and contains a Schrader-type valve, which allows the switch to be serviced without discharging the refrigerant system.

OPERATION

The A/C low pressure switch monitors the pressure of the refrigerant leaving the accumulator to the A/C compressor. When equipped with the 4.0L engine, the A/C low pressure switch is electrically connected in series with the A/C high pressure switch, between ground and the powertrain control module (PCM).

The contacts in the A/C low pressure switch open or close the path to ground, signaling the PCM to turn the A/C compressor clutch on and off. This regulates the refrigerant system pressure and controls A/C evaporator temperature. Controlling the evaporator temperature prevents condensate water on the evaporator fins from freezing and obstructing A/C system air flow.

The A/C low pressure switch contacts are open when the A/C suction line pressure is approximately 172 kPa (25 psi) or lower. The A/C low pressure switch contacts will close when the A/C suction line pressure rises to approximately 276 kPa (40 psi) or above. Lower ambient temperatures, below approximately -1° C (30° F), will also cause the A/C low pressure switch contacts to open. This is due to the pressure/temperature relationship of the refrigerant in the system.

The A/C low pressure switch is factory-calibrated and cannot be adjusted or repaired. If the A/C low pressure switch is faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

A/C LOW PRESSURE SWITCH

NOTE: Lower ambient temperatures, below about -1° C (30° F), during cold weather will open the A/C low pressure switch contacts and prevent A/C compressor operation due to the pressure/temperature relationship of the refrigerant.

Before performing diagnosis of the A/C low pressure switch, verify that the refrigerant system has the correct refrigerant charge (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE) and that the A/C low pressure switch is properly installed on the accumulator fitting. If the A/C low pressure switch is not properly installed, it may not open the Schrader-type valve in the accumulator fitting, which will prevent the switch from correctly monitoring the refrigerant system pressure.

For circuit descriptions and diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

(1) With gear selector in Park or Neutral and with park brake set, start the engine and allow it to idle.

(2) Place the A/C-heater mode control in any A/C position.

(3) Disconnect the wire harness connector from the A/C low pressure switch.

(4) Using a suitable jumper wire, install the jumper wire between the two terminals of the wire harness connector for the A/C low pressure switch. The A/C compressor clutch should engage. If the A/C clutch does engage, replace the faulty A/C low pressure switch. If the A/C clutch does not engage, the A/C clutch, A/C clutch relay, A/C-heater control , A/C high pressure switch (4.0L engine application), A/C pressure transducer (2.4L engine application), PCM, fuses or related wiring circuits may be defective (Refer to 8 - ELECTRICAL/WIRING DIAGRAM INFORMATION - DIAGNOSIS AND TESTING). Repair as required.